

FileEditSelectionViewGoRunTerminalHelp

3rd yr week 3 module

Update

EXPLORER

OPEN EDITORS

merge_sort.java

3RD YR WEEK 3 MODULE

merge_sort.class

merge_sort.java

merge_sort.java

merge_sort

public class merge_sort {
static void merge(int arr[], int left, int mid, int right) {
k++;
}
static void mergeSort(int arr[], int left, int right) {
if (left < right) {
int mid = (left + right) / 2;
mergeSort(arr, left, mid);
mergeSort(arr, mid + 1, right);
merge(arr, left, mid, right);
}
}
Run [Debug]
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int arr[] = new int[n];
for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
}
mergeSort(arr, left: 0, n - 1);
System.out.println(x: "Sorted Array:");
for (int i = 0; i < n; i++) {
System.out.print(arr[i] + " ");
}
}
}

CHAT

TERMINAL

uile>

cd "C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module\" ; if (\$?) { javac merge_sort.java } ; if (\$?) { java merge_sort }
6
450
1200
800
650
300
1500
Sorted Array:
300 450 650 800 1200 1500
PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module>

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.
+ merge_sort.java
Describe what to build
+ Agent Auto
Local Default Approvals

OUTLINE

TIMELINE

JAVA PROJECTS

Launchpad

Run Testcases

Live Share

Java: Ready

Ln 3, Col 26

Spaces: 4

UTF-8

CRLF

Java

18:20

07-07-2026

File Edit Selection View Go Run Terminal Help 3rd yr week 3 module

EXPLORER

- OPEN EDITORS
 - merge_sort.java 1
 - merge_sort_2.java 2
- 3RD YR WEEK 3 MODULE
 - merge_sort.class
 - merge_sort_2.class
 - merge_sort.java 1

merge_sort_2.java 2

```
1 public class merge_sort_2 {
2
34
35     Run | Debug
36     public static void main(String[] args) {
37
38         Scanner sc = new Scanner(System.in);
39
40         int n = sc.nextInt();
41
42         int arr[] = new int[n];
43
44         int sum = 0;
45
46         for (int i = 0; i < n; i++) {
47             arr[i] = sc.nextInt();
48             sum += arr[i];
49         }
50
51         mergeSort(arr, left: 0, n - 1);
52
53         System.out.println("Sorted Orders:");
54
55         for (int i = 0; i < n; i++) {
56             System.out.print(arr[i] + " ");
57         }
58
59         System.out.println();
60
61         int median;
62
63         if (n % 2 == 0)
64             median = (arr[n / 2 - 1] + arr[n / 2]) / 2;
65         else
66             median = arr[n / 2];
67
68         int count = 0;
69
70         for (int i = 0; i < n; i++) {
71             if (arr[i] > median)
72                 count++;
73         }
74     }
75 }
```

CHAT

TERMINAL

```
ule>
...
cd "C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module\" ; if ($?) { javac merge_sort_2.java } ; if ($?) { java merge_sort_2 }
7
12
25
18
30
15
22
40
Sorted Orders:
12 15 18 22 25 30 40
Median Order Value: 22
Orders Above Median: 3
Difference Between Highest and Lowest: 28
PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module>
```

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

- + merge_sort_2.java
- + Agent Auto

Local Default Approvals

Ln 100, Col 2 Spaces: 4 UTF-8 CRLF [] Java

18:24 07-07-2026

FileEditSelectionViewGoRunTerminalHelp

3rd yr week 3 module

Update

EXPLORER

OPEN EDITORS

merge_sort.java1

merge_sort_2.java2

Main.java1

3RD YR WEEK 3 MODULE

Main.class

Main.java1

merge_sort_2.class

merge_sort_2.java2

merge_sort.class

merge_sort.java1

OUTLINE

TIMELINE

JAVA PROJECTS

Launchpad

Run Testcases

Live Share

Java: Ready

Main.javaMain

public class Main {

static int partition(int arr[], int low, int high) {

int pivot = arr[high];

int i = low - 1;

for (int j = low; j < high; j++) {

if (arr[j] <= pivot) {

i++;

int temp = arr[i];

arr[i] = arr[j];

arr[j] = temp;

}

int temp = arr[i + 1];

arr[i + 1] = arr[high];

arr[high] = temp;

return i + 1;

}

static void quickSort(int arr[], int low, int high) {

if (low < high) {

int pi = partition(arr, low, high);

quickSort(arr, low, pi - 1);

quickSort(arr, pi + 1, high);

}

}

Run | Debug

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int arr[] = new int[n];

for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

}

}

CHAT

TERMINAL

CHAT SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ Main.java

+ Agent Auto

LocalDefault Approvals

Ln 60, Col 2

Spaces: 4

UTF-8

CRLF

Java

18:26

07-07-2026

```
uile> cd "c:\Users\sachi\Desktop\collage_module\3rd yr week
3 module\" ; if ($?) { javac Main.java } ; if ($?) { java
Main }
6
120
45
78
200
60
95
Sorted Array:
45 60 78 95 120 200
PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 mod
ule>
```

File Edit Selection View Go Run Terminal Help

3rd yr week 3 module

Update

EXPLORER

OPEN EDITORS

merge_sort.java 1

merge_sort_2.java 2

Main.java 1

quick_2.java

3RD YR WEEK 3 MODULE

Main.class

Main.java 1

merge_sort_2.class

merge_sort_2.java 2

merge_sort.class

merge_sort.java 1

quick_2.java

OUTLINE

TIMELINE

JAVA PROJECTS

Launchpad

Run Testcases

0 4

Live Share

Java: Ready

sort.java 1

merge_sort_2.java 2

Main.java 1

quick_2.java

quick_2.java

quick_2 > main(String[] args)

public class quick_2 {

Run/Debug

public static void main(String[] args) {}

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int arr[] = new int[n];

int sum = 0;

for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

sum += arr[i];

}

quickSort(arr, 0, n - 1);

System.out.println("Sorted Sales (Descending):");

for (int i = 0; i < n; i++) {

System.out.print(arr[i] + " ");

}

System.out.println();

int top = Math.min(n, 5);

int topSum = 0;

System.out.print("Top 5 Sales: ");

for (int i = 0; i < top; i++) {

System.out.print(arr[i] + " ");

topSum += arr[i];

}

System.out.println();

double topAverage = (double) topSum / top;

double overallAverage = (double) sum / n;

int count = 0;

for (int i = 0; i < n; i++) {

if (arr[i] > overallAverage) {

count++;

}

}

System.out.printf("Average of Top 5 Sales: %.2f\n", topAverage);

System.out.printf("Overall Average Sales: %.2f\n", overallAverage);

CHAT

TERMINAL

CHAT SESSIONS

Terminal

uile> cd "c:\Users\sachi\Desktop\collage_module\3rd yr week 3 module\" ; if (\$?) { javac Main.java } ; if (\$?) { java Main }

6

120

45

78

200

60

95

Sorted Array:

45 60 78 95 120 200

PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module>

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ quick_2.java

+ Agent Auto

Local Default Approvals

Ln 97, Col 5

Spaces: 4

UTF-8

CRLF

Java

18:29

07-07-2026

File Edit Selection View Go Run Terminal Help

3rd yr week 3 module

EXPLORER

- OPEN EDITORS
 - merge_sort.java
 - merge_sort_2.java
 - Main.java
 - quick_2.java
 - heap.java
- 3RD YR WEEK 3 MODULE
 - heap.java
 - Main.class
 - Main.java
 - merge_sort_2.class
 - merge_sort_2.java
 - merge_sort.class
 - merge_sort.java
 - quick_2.java
- OUTLINE
- TIMELINE
- JAVA PROJECTS

merge_sort_2.java 2

heap.java

```
1 public class heap {
2     static void heapify(int arr[], int n, int i) {
3         largest = i;
4
5         if (largest != i) {
6             int temp = arr[i];
7             arr[i] = arr[largest];
8             arr[largest] = temp;
9             heapify(arr, n, largest);
10        }
11    }
12
13    static void heapSort(int arr[]) {
14        int n = arr.length;
15        for (int i = n / 2 - 1; i >= 0; i--) {
16            heapify(arr, n, i);
17        }
18        for (int i = n - 1; i > 0; i--) {
19            int temp = arr[0];
20            arr[0] = arr[i];
21            arr[i] = temp;
22            heapify(arr, i, 0);
23        }
24    }
25
26    public static void main(String[] args) {
27        Scanner sc = new Scanner(System.in);
28
29        int n = sc.nextInt();
30
31        int arr[] = new int[n];
32        for (int i = 0; i < n; i++) {
33            arr[i] = sc.nextInt();
34        }
35
36        heapSort(arr);
37
38        System.out.println("Sorted Array:");
39        for (int i = 0; i < n; i++) {
40            System.out.print(arr[i] + " ");
41        }
42        sc.close();
43    }
44}
```

Run | Debug

CHAT

Terminal

```
PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module> cd "c:\Users\sachi\Desktop\collage_module\3rd yr week 3 module\"; if ($?) { javac Main.java }; if ($?) { java Main }
6
120
45
78
200
60
95
Sorted Array:
45 60 78 95 120 200
PS C:\Users\sachi\Desktop\collage_module\3rd yr week 3 module>
```

CHAT SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ heap.java

+ Agent Auto

Local Default Approvals

Ln 32, Col 10 Spaces: 4 UTF-8 CRLF Java

18:30 07-07-2026

File Edit Selection View Go Run Terminal Help

3rd yr week 3 module

Update

EXPLORER

OPEN EDITORS

merge_sort.java 1

merge_sort_2.java 2

Main.java 1

quick_2.java

heap.java 1

heap_2.java

3RD YR WEEK 3 MODULE

heap_2.java

heap.java 1

Main.class

Main.java 1

merge_sort_2.class

merge_sort_2.java 2

merge_sort.class

merge_sort.java 1

quick_2.java

OUTLINE

TIMELINE

JAVA PROJECTS

merge_sort.java 1

merge_sort_2.java 2

Main.java 1

quick_2.java

heap.java 1

heap_2.java

heap_2.java

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public class heap_2 {

static void heapSort(int arr[]) {

int n = arr.length;

for (int i = n / 2 - 1; i >= 0; i--) {

heapify(arr, n, i);

}

for (int i = n - 1; i > 0; i--) {

int temp = arr[0];

arr[0] = arr[i];

arr[i] = temp;

heapify(arr, i, i);

}

}

Run|Debug

public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

int n = sc.nextInt();

int arr[] = new int[n];

int sum = 0;

for (int i = 0; i < n; i++) {

arr[i] = sc.nextInt();

sum += arr[i];

}

heapSort(arr);

System.out.println("Sorted Response Times:");

for (int i = 0; i < n; i++) {

System.out.print(arr[i] + " ");

}

System.out.println();

int fastest = arr[0];

int slowest = arr[n - 1];

double average = (double) sum / n;

int count = 0;

for (int i = 0; i < n; i++) {

if (arr[i] < average) {

count++;

}

}

}

CHAT

TERMINAL

CHAT

SESSIONS

Tip: Try the Plan agent to research and plan before implementing changes.

+ heap_2.java

Describe what to build

+ | | Auto | |

Lo... | Default Appo...

Ln 34, Col 10

Spaces: 4

UTF-8

CRLF

Java

18:44

07-07-2026

```
import java.util.Scanner;
```

```
public class Main {
```

```
static void merge(int arr[], int left, int mid,
```

```
int n1 = mid - left + 1;
```

```
int n2 = right - mid;
```

```
int L[] = new int[n1];
```

```
int R[] = new int[n2];
```

```
for (int i = 0; i < n1; i++)
```

```
L[i] = arr[left + i];
```

```
for (int j = 0; j < n2; j++)
```

```
R[j] = arr[mid + 1 + j];
```

```
int i = 0, j = 0, k = left;
```

```
while (i < n1 && j < n2) {
```

```
if (L[i] <= R[j]) {
```

```
arr[k] = L[i];
```

```
i++;
```

```
} else {
```

```
arr[k] = R[j];
```

```
}
```

```
}
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
int arr[] = new int[n];
```

```
for (int i = 0; i < n; i++) {
```

```
arr[i] = sc.nextInt();
```

```
}
```

```
mergeSort(arr, 0, n - 1);
```

```
System.out.println("Sorted Array:");
```

```
for (int i = 0; i < n; i++) {
```

```
System.out.print(arr[i] + " ");
```

```
}
```

```
}
```

```
}
```

```
arr[0] = arr[i];  
arr[i] = temp;
```

```
heapify(arr, i, 0);  
}  
}
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
int arr[] = new int[n];
```

```
int sum = 0;
```

```
for (int i = 0; i < n; i++) {  
arr[i] = sc.nextInt();  
sum += arr[i];  
}
```

```
heapSort(arr);
```

```
System.out.println("Sorted Response Times:")
```

```
int count = 0;
```

```
for (int i = 0; i < n; i++) {  
    if (arr[i] < average) {  
        count++;  
    }  
}
```

```
double percentage = (double) count * 100 / n;
```

```
System.out.println("Fastest: " + fastest);  
System.out.println("Slowest: " + slowest);  
System.out.printf("Average: %.2f\n", average);  
System.out.println("Cases Faster Than Average: " + count);  
System.out.printf("Percentage: %.2f%%", percentage);
```

```
sc.close();  
}
```

```
}
```

```
import java.util.Scanner;  
public class heap_2 {
```

```
// Heapify Function
```

```
static void heapify(int arr[], int n, int i) {
```

```
    int largest = i;
```

```
    int left = 2 * i + 1;
```

```
    int right = 2 * i + 2;
```

```
    if (left < n && arr[left] > arr[largest]) {  
        largest = left;  
    }
```

```
    if (right < n && arr[right] > arr[largest]) {  
        largest = right;  
    }
```

```
    if (largest != i) {
```

```
        int temp = arr[i];
```

```
        arr[i] = arr[largest];
```

```
        arr[largest] = temp;
```

```
        heapify(arr, n, largest);  
    }
```

```
import java.util.Scanner;  
public class heap {  
    static void heapify(int arr[], int n, int i) {
```

```
        int largest = i;  
        int left = 2 * i + 1;  
        int right = 2 * i + 2;
```

```
        if (left < n && arr[left] > arr[largest]) {  
            largest = left;  
        }
```

```
        if (right < n && arr[right] > arr[largest]) {  
            largest = right;  
        }
```

```
        if (largest != i) {
```

```
            int temp = arr[i];  
            arr[i] = arr[largest];  
            arr[largest] = temp;
```

```
            heapify(arr, n, largest);  
        }  
    }
```

```
int arr[] = new int[n];
```

```
for (int i = 0; i < n; i++) {  
    arr[i] = sc.nextInt();  
}
```

```
heapSort(arr);
```

```
System.out.println("Sorted Array:");
```

```
for (int i = 0; i < n; i++) {  
    System.out.print(arr[i] + " ");  
}
```

```
sc.close();  
}  
}}
```

```
import java.util.Scanner;
```

```
public class quick_2 {
```

```
// Partition for Descending Order
```

```
static int partition(int arr[], int low, int high)
```

```
{  
    int pivot = arr[high];
```

```
    int i = low - 1;
```

```
    for (int j = low; j < high; j++) {
```

```
        if (arr[j] >= pivot) {
```

```
            i++;
```

```
            int temp = arr[i];
```

```
            arr[i] = arr[j];
```

```
            arr[j] = temp;
```

```
        }
```

```
    }
```

```
    int temp = arr[i + 1];
```

```
    arr[i + 1] = arr[high];
```

```
    arr[high] = temp;
```

```
}
```

```
}
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
int arr[] = new int[n];
```

```
int sum = 0;
```

```
for (int i = 0; i < n; i++) {
```

```
arr[i] = sc.nextInt();
```

```
sum += arr[i];
```

```
}
```

```
quickSort(arr, 0, n - 1);
```

```
System.out.println("Sorted Sales (Descendin
```

```
for (int i = 0; i < n; i++) {
```

```
System.out.print(arr[i] + " ");
```

```
}
```

```
}
```

```
System.out.println();
```

```
double topAverage = (double) topSum / top;
```

```
double overallAverage = (double) sum / n;
```

```
int count = 0;
```

```
for (int i = 0; i < n; i++) {  
    if (arr[i] > overallAverage) {  
        count++;
```

```
    }
```

```
}
```

```
System.out.printf("Average of Top 5 Sales: %2f\n",  
    topAverage);  
System.out.printf("Overall Average Sales: %2f\n",  
    overallAverage);
```

```
System.out.println("Sales Above Overall Average: " + count);
```

```
sc.close();
```

```
}
```

```
}
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    static int partition(int arr[], int low, int high)
```

```
    {
        int pivot = arr[high];
```

```
        int i = low - 1;
```

```
        for (int j = low; j < high; j++) {
```

```
            if (arr[j] <= pivot) {
```

```
                i++;
```

```
                int temp = arr[i];
```

```
                arr[i] = arr[j];
```

```
                arr[j] = temp;
```

```
            }
```

```
        }
```

```
        int temp = arr[i + 1];
```

```
        arr[i + 1] = arr[high];
```

```
        arr[high] = temp;
```

```
        return i + 1;
```

```
}
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
int n = sc.nextInt();
```

```
int arr[] = new int[n];
```

```
for (int i = 0; i < n; i++) {
```

```
arr[i] = sc.nextInt();
```

```
}
```

```
quickSort(arr, 0, n - 1);
```

```
System.out.println("Sorted Array:");
```

```
for (int i = 0; i < n; i++) {
```

```
System.out.print(arr[i] + " ");
```

```
}
```

```
}
```

```
}
```

```
import java.util.Scanner;  
public class merge_sort_2 {
```

```
    static void merge(int arr[], int left, int mid,
```

```
        int n1 = mid - left + 1;
```

```
        int n2 = right - mid;
```

```
        int L[] = new int[n1];
```

```
        int R[] = new int[n2];
```

```
        for (int i = 0; i < n1; i++)
```

```
            L[i] = arr[left + i];
```

```
        for (int j = 0; j < n2; j++)
```

```
            R[j] = arr[mid + 1 + j];
```

```
        int i = 0, j = 0, k = left;
```

```
        while (i < n1 && j < n2) {
```

```
            if (L[i] <= R[j]) {
```

```
                arr[k] = L[i];
```

```
                i++;
```

```
            } else {
```

```
                arr[k] = R[j];
```

```
                j++;
```

```
public static void main(String[] args) {  
    Scanner sc = new Scanner(System.in);  
  
    int n = sc.nextInt();  
  
    int arr[] = new int[n];  
  
    int sum = 0;  
  
    for (int i = 0; i < n; i++) {  
        arr[i] = sc.nextInt();  
        sum += arr[i];  
    }  
  
    mergeSort(arr, 0, n - 1);  
  
    System.out.println("Sorted Orders:");  
  
    for (int i = 0; i < n; i++) {  
        System.out.print(arr[i] + " ");  
    }  
  
    System.out.println();  
}
```